E-Commerce US Dataset Analysis

Report 7: Business Insights and Recommendations

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1. Revenue Summary

- Total revenue from the 10,000-order sample: approximately $12.09 million.
- Average order value (AOV): $1,209.
- Top 3 revenue-generating states (CA, TX, FL) drive 57% of total revenue.
- Electronics category alone accounts for 38.6% of item volume — the single largest revenue driver.

2. Customer Insights

2.1 Repeat Purchase Behavior

- 21% of customers placed more than one order.
- Repeat customers generate a disproportionate share of revenue relative to their headcount.
- VIP segment (9.8% of customers) is the retention-critical group to monitor and invest in.

2.2 Segment Behavior

Segment	Share	Behavior Profile
Consumer	69.8%	High volume, lower individual order values; price-sensitive
Corporate	20.5%	Larger basket sizes; likely buying for teams or business use
VIP	9.8%	Highest spend; strong review engagement; highest repeat rate

2.3 Demographic Profile

- Core buying demographic: women aged 35–55 in CA, TX, and FL.
- Male buyers (35.7%) skew toward Electronics; female buyers (64.3%) spread across all categories.

3. Product Insights

3.1 Category Concentration Risk

- Electronics (38.6% of items) creates single-category dependency.
- Fashion (20%) and Books (14.5%) are growing categories with room to expand SKU depth.
- Home Goods, Auto, Furniture, and Toys each sit below 8% — underpenetrated relative to market demand.

3.2 Pricing Tiers

- Mainstream tier ($100–$500): highest volume, strongest conversion.
- Premium tier (>$1,746): 4% of items; higher margin but slower turnover.
- Cost-to-price data available for all products — margin analysis by category is feasible.

4. Seller Insights

4.1 Revenue Concentration

- 72% of sellers (362 out of 500) generate 80% of total revenue.
- 28% of sellers (138) share the remaining 20% of revenue — low contribution and low growth signal.
- Top seller significantly outpaces all others — platform has single-seller concentration risk.

4.2 Geographic Alignment

- Seller locations mirror buyer locations in CA, TX, and NY.
- Strong demand-supply geographic overlap reduces shipping times and freight costs.

5. Operational Insights

5.1 Delivery Performance

Metric	Value
Delivered orders	93.3% (9,332)
Canceled orders	6.7% (668)
On-time delivery rate	90.4%
Average delivery time	7.5 days
Late delivery rate	9.6%

- On-time delivery (90.4%) is the primary driver of positive reviews — protecting it is a business priority.
- The 9.6% late delivery rate links directly to the lower review scores (~3.6 avg for slow orders).

5.2 Review Satisfaction

- Overall avg review score: 4.05/5.
- Fast deliveries (<5 days) average 4.3 stars; slow deliveries (>10 days) average 3.6 stars.
- Investment in logistics speed has a quantifiable payoff in customer rating.

6. Payment Insights

- Credit card is the dominant payment method (50.3%) — supports both single and installment payments.
- 54% of orders use installments — deferred payments are enabling order completion, especially for Electronics.
- Apple Pay at 8.6% signals a growing mobile-first buyer segment worth tracking.
- Boleto (2.5%) is the smallest method — may be a legacy channel with declining relevance.

7. Recommendations

7.1 Geographic Expansion

- CO, WA, and AZ each account for less than 3% of orders despite having major metro populations.
- Targeted marketing in Denver, Seattle, and Phoenix represents a low-risk growth play.
- Consider seller recruitment in underserved states to reduce freight distance and delivery time.

7.2 Category Diversification

- Reduce revenue dependency on Electronics by deepening Fashion and Home Goods catalogs.
- Expand Books SKU selection — stable demand (14.5%) with low return rates typical in this category.
- Run cross-category promotions to introduce Electronics buyers to Furniture and Home Goods.

7.3 Seller Performance Program

- Introduce performance tiers with volume-based commission incentives for Top and Mid sellers.
- Provide bottom-tier sellers (28%) with tools or coaching to improve product listing quality and fulfillment speed.
- Investigate the 7 multi-brand sellers — they carry operational leverage but also concentration risk.

7.4 Customer Retention

- Build a loyalty program targeting the VIP segment (9.8% of customers, disproportionate revenue share).
- One-time buyers (79% of customer base) are a conversion opportunity — trigger re-engagement campaigns at 30/60/90 days post-order.

- Use engagement_score from customer_features.csv to prioritize outreach budgets.

7.5 Delivery Improvement

- Late deliveries (9.6%) concentrate in Slow delivery speed tier — investigate seller-level and region-level patterns.
- Set a short-term target to reduce late rate from 9.6% to under 7%.
- Share real-time delivery data with sellers to allow proactive intervention before the estimated date passes.

7.6 Payment Strategy

- Protect installment payment options — 54% uptake shows they are material to order completion.
- Monitor Apple Pay and PayPal growth; ensure mobile checkout flows are optimized.
- Consider targeted voucher campaigns for Corporate segment to increase basket size.

8. Modeling Opportunities

- Churn prediction: 79% one-time buyers; identify characteristics distinguishing repeaters from one-timers.
- Late delivery prediction: model delivery_delay using seller_tier, product_weight_g, delivery_speed, and customer_state.
- Price optimization: cost-to-price data available for all 2,000 SKUs; build margin-sensitivity models by category.
- Product recommendation: 10,000 orders across 2,000 products — sufficient for collaborative filtering baseline.
- Customer segmentation: engagement_score and value_tier are ready-made inputs for RFM-style clustering.